Black-Box Modelling of Multiband Saturation

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Abstract

Virtual analog modelling has become an active area of research as musicians and producers seek software alternatives to expensive and inaccessible hardware processors. Multiband saturators present a particularly demanding emulation target: they combine frequency-dependent nonlinear distortion across several bands with phase-coherent crossover filtering. This work addresses black-box modelling of multiband saturation using paired electric bass recordings from the IDMT-SMT-Bass dataset. We compare a bidirectional LSTM and a WaveNet-style convolutional network trained with a combined time-domain and multi-resolution spectral loss.

A supplementary experiment confirms that emulating a multiband configuration is harder than single-band saturation, even when the same nonlinear style is applied across all bands.

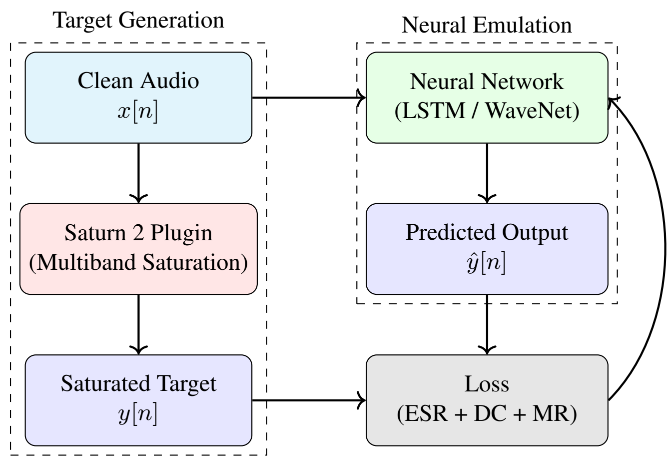

Figure 1. *Overview of the proposed emulation pipeline. Clean bass audio is processed through a Saturn 2 preset to generate supervised targets. A deep neural network (LSTM or WaveNet) is trained to approximate the nonlinear behavior by minimizing the error between predicted and target waveforms.*

1. Introduction

Digital audio effects modelling is an active area of research aiming to **replicate the behavior of analog audio processors digitally**, allowing bulky, expensive, and fragile hardware units to be replaced by **accessible software solutions** running on a modern desktop computer.

From a Deep Learning perspective, saturation is one of the most compelling categories of audio effects, unlike linear processors such as equalizers or reverbs, saturators introduce **nonlinear harmonic distortion** that cannot be captured by conventional linear filters, making them a perfect target for neural network modelling.

Due to the lack of hardware access, the target system for this project shifted from hardware to software, while still preserving the original purpose of emulating the complex nonlinear audio transformation of the saturation. The target software is the FabFilter Saturn 2 VST plugin [1], which also allowed us to configure a **Multiband Saturation Preset**, making it a particularly demanding and interesting emulation target.

The task is formulated as supervised waveform-to-waveform regression: given a clean electric bass DI (direct input) signal as input, the model must predict the corresponding saturated output produced by the plugin, with no access to the plugin's internal processing code, hence the term **black-box modelling** [2].

A common approach for virtual analog modelling of distortion effects is **"white-box" modeling** that simulates analog circuits based on their component analysis. While highly accurate when circuit details are known, it is computationally demanding and labor-intensive for complex systems. Alternatively, black-box modeling bypasses circuit analysis entirely, creating a model that replicates the observed input-output mapping based **solely on response measurements.**

The central question we address is: *Can a deep neural network accurately emulate multiband saturation from clean bass input alone, and which model, LSTM or WaveNet, achieves better emulation fidelity, under matched conditions?*

Since the primary goal for this project is the **emulation accuracy** (assessed through both signal-domain metrics and qualitative listening evaluation), **offline processing** is preferred. Therefore, **real-time use** is not a priority and falls outside the scope of this work.

The source code repository, along with an interactive audio demonstration page containing the model predictions, is publicly available online.[1]

2. Dataset and Saturation Pipeline

2.1. Source dataset

The foundation of our data is the **IDMT-SMT-Bass dataset (Fraunhofer IDMT)** [3], which contains approximately 5,200 WAV files (44.1 kHz, 24-bit) of single recorded electric bass notes, covering the range E1 (41.2 Hz) to G3 (196.0 Hz). All recordings are captured as DI signals, taken directly from the instrument output, with no effects processing applied, making them ideal clean inputs for our supervised learning setup. The dataset includes three different 4-string electric bass guitars and includes ten playing techniques, namely five plucking styles and five expression styles.

2.2. Dataset filtering

We restrict the original dataset to four plucking styles: Fingerstyle (FS), Muted (MU), Picked (PK), and Slap-Thumb (ST). Expression styles involving pitch modulation such as vibrato, bending, harmonics and slides are excluded, as they introduce frequency variation unrelated to the nonlinear behaviour of saturation. Slap-Pluck technique is also excluded due to its high transient amplitude, causing excessive and inconsistent distortion that biases the model and degrades generalization performance. The retained combinations provide a **controlled range** of attack transients and timbral characteristics, from the smooth sustain of fingerstyle to the (relatively) aggressive percussive transient of slap-thumb.

2.3. Saturation pipeline

Supervised training targets were generated by batch-processing keyword-filtered WAV files from the IDMT-SMT-Bass directory through the FabFilter Saturn 2 VST3 plugin. Using the **Spotify Pedalboard** Python library, we loaded the plugin and saved the resulting saturated signals to a mirrored output directory with a "_saturated" suffix in each filename. This structure organizes the final dataset into clean and saturated folders with directly corresponding filenames.

Additionally, a YAML file stores deterministic configuration parameters such as the IDMT-SMT-Bass dataset path, the output path (where the processed dataset will be generated), the VST3 plugin path and filters containing the desired keywords to the corresponding audio files to include. This design, besides being user-friendly, **minimizes the need for modifications to the main saturation pipeline**, as all relevant information is automatically extracted from the YAML file.

2.4. Preset configuration

The Saturn 2 preset used, is actually a stock preset called *"Power Overdrive I - 4 Bands SM"*, which consists of a **four band configuration**, using the Power Amp saturation style across all bands, although **applied differently in each one**. The crossover frequencies are set at 61.2 Hz, 249.3 Hz, and 2506.0 Hz, with slopes of 24 dB/oct. Drive levels range from 31.8% in the low-mid band to 53.5% in the sub band, with per-band tone shaping applied. However, **all dynamic parameters involved** (envelope followers, feedback) **were set to neutral or zero**, ensuring that the target transformation remains time-invariant for a given input amplitude. The mix is fixed at 100% (fully wet) across all bands, and the master output gain is set to 0.0 dB.

Figure 2. *FabFilter Saturn 2 VST3 Plugin with the Power Overdrive I - 4 Bands SM preset configuration.*

All knob parameters are also stored in a YAML file, since the Spotify Pedalboard library can't load the original preset file. However, this approach also enables practical preset management and easy **integration of additional external presets**, as the saturation pipeline automatically extracts and applies these parameters from the YAML files.

2.5. Signal Analysis

Figure 3 compares a representative clean and saturated audio file. The saturation introduces visible harmonic enrichment across the spectrum, confirming the nonlinear nature of the transformation the model must learn.

This processing revealed a peak amplitude increase from 0.3197 to 0.5634, and a mean RMS energy rise from 0.0576 to 0.2124, corresponding to an average RMS gain factor of

[1]Source Code: https://github.com/joao-canais/Black-Box-Modelling-of-Multiband-Saturation

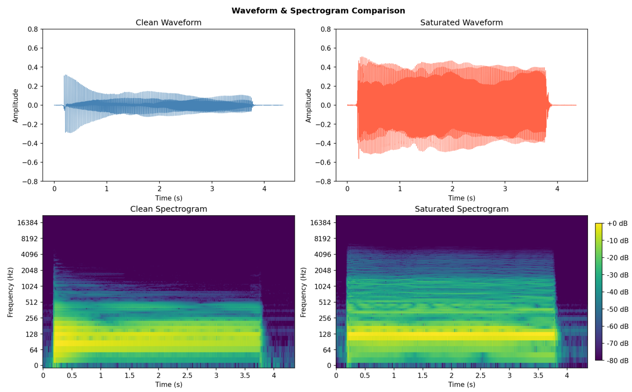

Figure 3. *Waveform (top) and Spectrogram (bottom) of a bass note before and after Saturn 2 processing.*

approximately 3.69×. Both signals remain within the safe digital range (below 1.0), confirming that no hard clipping was introduced during rendering.

A latency of 292 samples (6.62 ms) was also estimated between the clean and saturated signals. This delay is an intrinsic consequence of the plugin's linear-phase crossover filters and was **intentionally preserved** in the dataset (as explained in more detail in Section 3.1).

2.6. Dataset Split Strategy

Since the dataset includes recordings from three distinct instruments, a random split could lead to optimistic results due to timbral leakage across sets. The model **might learn the instrument characteristics** instead of the intended saturation mapping.

To mitigate this, we adopted a **1-1-1 Split** where each instrument is exclusively assigned to one subset (train, validation, or test). This enforces complete timbral separation, providing a strict evaluation of generalization.

Table 1. *Dataset Distribution and Duration Statistics by Split and Plucking Style*

Dataset Split	Plucking Style	File Count	Duration
Train (Guitar 1)	Finger Style (FS)	156	9m 51s
Subtotal: 624 pairs	Muted (MU)	156	6m 32s
Duration: 33m 54s	Picked (PK)	156	8m 54s
	Slap-Thumb (ST)	156	8m 37s
Val. (Guitar 2)	Finger Style (FS)	30	1m 25s
Subtotal: 134 pairs	Muted (MU)	32	1m 14s
Duration: 5m 31s	Picked (PK)	37	1m 36s
	Slap-Thumb (ST)	35	1m 17s
Test (Guitar 3)	Finger Style (FS)	39	1m 47s
Subtotal: 134 pairs	Muted (MU)	37	1m 22s
Duration: 5m 34s	Picked (PK)	31	1m 22s
	Slap-Thumb (ST)	27	1m 03s
Total Dataset Duration			**44m 59s**

3. Models

3.1. Long Short-Term Memory (LSTM)

Although our baseline model follows the architecture proposed by Wright et al. [4, 5] (designed for real-time use), in this work, we adapted it for offline emulation by replacing the causal LSTM with a **two-layer Bidirectional LSTM**, followed by a **fully connected layer** that projects the hidden state $h[n]$ to a scalar output sample $\hat{y}[n]$. The hidden size is subjected to hyperparameter tuning, with values of 32, 64, and 96.

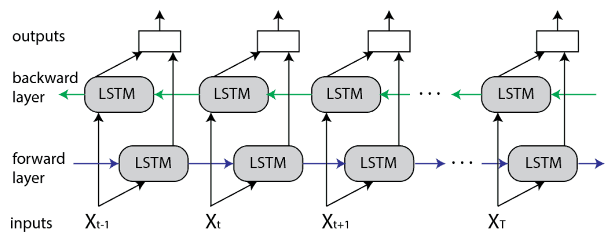

Figure 4. *Proposed Bidirectional LSTM model. The input signal $x[n]$ is processed by forward and backward LSTM layers whose hidden states are concatenated and projected to a scalar output sample $\hat{y}[n]$ via a fully connected layer.*

At each time step n, the model receives one input sample $x[n]$ and produces one predicted output sample $\hat{y}[n]$. Unlike a causal LSTM, the Bidirectional LSTM processes the full input segment in both forward and backward directions simultaneously, capturing both past and future temporal context. This design choice is motivated by the nature of the target plugin: the FabFilter Saturn 2 employs linear-phase crossover filters that introduce a deliberate look-ahead latency, meaning the saturated output at time t depends on future input samples $x[t+1], x[t+2], \ldots, x[t+k]$. Therefore, this latency was deliberately preserved in the dataset so that the model learns to replicate the complete behaviour of the target system. A strictly causal model cannot resolve this dependency, leading to systematic prediction errors and early divergence of the validation loss.

The hidden states of both directions are concatenated to form $h[n] \in \mathbb{R}^{2\times\text{hidden\_size}}$, which is then projected to a scalar output by the fully connected layer:

$$\hat{y}[n] = W_{fc} \cdot [h^{\rightarrow}[n] \oplus h^{\leftarrow}[n]] + b_{fc}$$

The target $y[n]$ is the corresponding sample from the Saturn 2 output and the prediction $\hat{y}[n]$ is the scalar produced by the fully connected layer on top of the concatenated forward and backward hidden states $[h^{\rightarrow}[n] \oplus h^{\leftarrow}[n]]$.

Also a **skip connection** (or residual connection) adds the input directly to the output of the linear layer, enabling the network to learn only the residual difference between the clean and saturated signals rather than the full output waveform. This design stabilises training and improves early convergence, as also reported in Wright et al. [5].

3.2. WaveNet Convolutional Neural Network

The WaveNet-style neural network [6] consists of a series of convolutional layers. The raw input waveform is given as input to the first convolutional layer. The convolutional layers apply linear filtering and a nonlinear activation function to the signal. The post-processing module is replaced by a linear mixer, a single linear 1 × 1 convolution layer. According to the experiments done by Damskägg et al. [7], the network performs similarly or better with the linear output layer.

3.2.1. Receptive Field

The proposed WaveNet-style model is causal and feedforward: the predicted output sample at time instant n depends only on the N most recent input samples, where N is the receptive field of the model. The receptive field is determined by the number of convolutional layers and the dilation factors applied at each layer, and is given by:

$$N = (M-1)\sum_{k=1}^{K} d_k + 1$$

where K is the total number of convolutional layers, M is the kernel size, and d_k is the dilation factor of layer k.

To illustrate this, consider Figure 5, where a network with $K = 3$ convolutional layers with dilation factors $d_k = \{1, 2, 4\}$ and kernel size $M = 2$.

Applying the formula:

$$N = (2-1) \times (1+2+4) + 1 = 8 \text{ samples}$$

meaning the output at time n depends on the 8 most recent input samples.

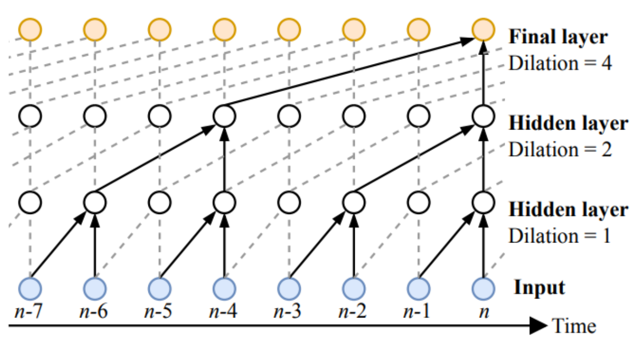

Figure 5. *Illustration of three convolutional layers in series and the resulting receptive field of N = 8.*

3.2.2. Gated Activation Functions

Following the original WaveNet formulation [6], the gated activation unit adopted in this work splits the convolutional output into two equal halves along the channel dimension and multiplies them element-wise:

$$z = \tanh(H_f * x) \odot \sigma(H_g * x)$$

where $*$ denotes a convolution operator, $\odot$ denotes an element-wise multiplication operator, $\sigma(\cdot)$ is the logistic sigmoid function, and H_f, H_g are the filter and gate convolution kernels respectively.

4. Training

4.1. Loss Function

To train the models robustly, we use a combined loss function with three factors from the **auraloss library** [8], addressing emulation fidelity in both the time and frequency domains.

4.1.1. ESR with Pre-emphasis

The Error-to-Signal Ratio (ESR) measures the squared error between the input $\hat{y}$ and target y, normalised by the energy of the target signal. However, bass guitar signals concentrate most energy in the low frequencies, where saturation has relatively little effect. To compensate, a **pre-emphasis filter** was applied to the output and target signals before computing the ESR in order to emphasise certain frequencies in the loss function such as the mid and high frequencies, since preliminary experiments in Wright et al.[5] indicated that without the pre-emphasis filtering the resulting neural network produced audible errors in the mid to high frequency range.

The pre-emphasis filter was chosen to be a first-order high-pass filter with transfer function:

$$H(z) = 1 - 0.85z^{-1} \longrightarrow y_p[n] = y[n] - 0.85 \cdot y[n-1]$$

For a segment of training signal of length N, the pre-emphasised ESR is given by:

$$\mathcal{E}_{\text{ESR}} = \frac{\sum_{n=0}^{N-1} |y_p[n] - \hat{y}_p[n]|^2}{\sum_{n=0}^{N-1} |y_p[n]|^2},$$

where y_p is the pre-emphasised target signal, and $\hat{y}_p$ is the preemphasised neural network output.

4.1.2. DC Loss

A lightweight DC loss term was also added [4, 5] to penalise any residual mean offset in the model output. It is given as the squared mean error divided by the squared mean of the target:

$$\mathcal{E}_{\text{DC}} = \frac{\left| \frac{1}{N} \sum_{n=0}^{N-1} (y_p[n] - \hat{y_p}[n]) \right|^2}{\frac{1}{N} \sum_{n=0}^{N-1} |y_p[n]|^2}.$$

4.1.3. Multi-Resolution STFT Loss (MRSTFT)

To complement the time-domain loss, we include a Multi-Resolution STFT[2] (MRSTFT) loss, which computes the spectral difference between the predicted and target signals simultaneously across multiple FFT window sizes [8]. This helps the model to reproduce the correct harmonic enrichment profile of the multiband saturation across different frequency resolutions, particularly important given the wide spectral footprint of the four-band preset.

$$\mathcal{E}_{\text{MR}} = \frac{1}{M} \sum_{m=1}^{M} (\mathcal{E}_{\text{SC}}(\hat{y_p}, y_p) + \mathcal{E}_{\text{SM}}(\hat{y_p}, y_p)).$$

where M is the number of STFT resolutions, and $\mathcal{E}_{\text{SC}}$ and $\mathcal{E}_{\text{SM}}$ denote the spectral convergence and spectral log-magnitude losses for each resolution, defined as:

$$\mathcal{E}_{\text{SC}} = \frac{\| \, |\text{STFT}(y_p)| - |\text{STFT}(\hat{y_p})| \, \|_{\text{F}}}{\| \, |\text{STFT}(y_p)| \, \|_{\text{F}}}$$

$$\mathcal{E}_{\text{SM}} = \frac{1}{N} \| \log(|\text{STFT}(y_p)|) - \log(|\text{STFT}(\hat{y_p})|) \|_1$$

4.1.4. Combined Loss

The final training loss combines all three terms with respective weights:

$$\mathcal{E} = \mathcal{E}_{ESR} \cdot \lambda_{ESR} + \mathcal{E}_{DC} \cdot \lambda_{DC} + \mathcal{E}_{MR} \cdot \lambda_{MR}$$

With $\lambda_{ESR} = 0.45$, $\lambda_{DC} = 0.1$, $\lambda_{MR} = 0.45$. The equal weighting of ESR and MRSTFT reflects their complementary roles, with ESR focusing on time-domain accuracy and MRSTFT capturing spectral structure across multiple resolutions.

4.2. Batch-Processing

For the LSTM, audio segments are reshaped into 3D tensors of shape $[batch\_size,\ seg\_len,\ input\_size]$, where $input\_size = 1$ since the input is a mono audio signal. For the WaveNet, the segments follow the standard PyTorch 1D convolution convention, adopting the shape $[batch\_size,\ 1,\ seg\_len]$, where the channel dimension precedes the temporal dimension.

Both models use a segment length of 22050 samples and are trained with a batch size of 16, using the Adam optimizer with an initial learning rate of 5×10^{-4}.

Validation is computed every 2 training epochs and training terminates early if the validation loss does not improve for 25 consecutive validation checks, which is equivalent to 50 training epochs without improvement [2].

[2] Short-time Fourier Transform

5. Results

Both models were trained on shared GPU servers provided by Faculdade de Ciências da Universidade de Lisboa (FCUL), using whichever was available at the time:

- *Opel* (2× AMD EPYC 7443, 96 hyper-threads, 128 GB RAM, 6× NVIDIA A30 24 GB)
- *Corsa* (2× Intel Xeon Silver 4216, 64 hyper-threads, 128 GB RAM, 2× NVIDIA Tesla T4 16 GB)

Access to this infrastructure allowed us to run extended training sessions with higher epoch budgets and to perform **hyperparameter tuning across both architectures.**

Table 2. *Test Set Evaluation for the different Proposed Models. The best result for each architecture is highlighted.*

Approach	Hidden Size	Layers	Number of Parameters	Total Loss $\mathcal{L}_{\text{total}}$
LSTM 1	32	2	34 113	0.40122
LSTM 2	64	2	133 761	0.38625
LSTM 3	96	2	298 945	**0.17684**
WaveNet 1	16	10	13 473	0.38758
WaveNet 2	16	18	24 225	**0.34709**

As shown in Table 2, the proposed Bidirectional LSTM architecture improves greatly with model capacity, dropping from $\mathcal{L}_{\text{total}} \approx 0.40$ at hidden size 32 to ≈ 0.18 at hidden size 96.

Although the WaveNet model also improved with the 18-layer configuration, the total loss remains well above the best LSTM result. Notably, the *WaveNet 2* achieves a lower total loss than *LSTM 2* while using roughly $5.5\times$ fewer parameters, suggesting that the WaveNet architecture is more parameter-efficient for this task.

This total loss gap between LSTM and WaveNet neural networks highlights the advantage of the Bidirectional LSTM since the FabFilter Saturn 2 employs linear-phase crossover filters whose group delay means that the saturated output at time t depends on input samples from both past and future instants, a bidirectional recurrent architecture is then better suited to capturing this non-causal dependency than a feedforward WaveNet, whose receptive field remains constrained to the past.

Larger model configurations, such as *LSTM 3*, inherently require **more epochs to converge due to their high number of parameters**. This is evident in our results since this specific LSTM, with 298,945 parameters, was trained for approximately 15 hours over 1000 epochs. From epoch 500 onward, performance gains became increasingly marginal, with the validation loss decreasing by 0.01. This stabilization aligns with the learning rate scheduler progressively scaling down the learning rate, indicating that the model

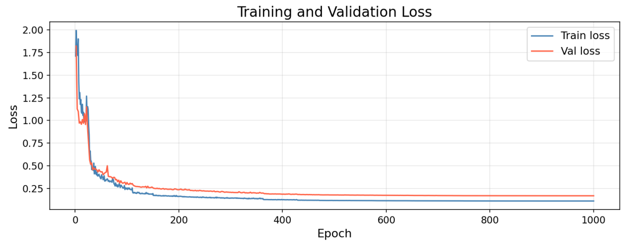

Figure 6. *Training and Validation combined loss curves for LSTM 3 over 1000 epochs (15h 2m). The best validation loss of 0.169 was achieved at epoch 992.*

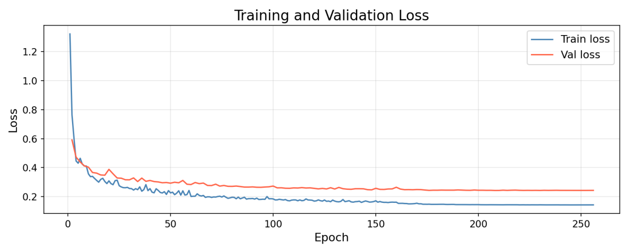

Figure 7. *Training and Validation combined loss curves for WaveNet 2 over 258 epochs (5h 58m). The best validation loss of 0.243 was achieved at epoch 208.*

had effectively reached its convergence limit.

In contrast, the *WaveNet 2* converged considerably faster, triggering early stopping at epoch 258 after approximately 6 hours of training. The earlier convergence reflects the feed-forward nature of the WaveNet architecture, which does not maintain a recurrent hidden state and therefore optimises more efficiently per epoch.

The quantitative evaluation on the test set is split into two main perspectives: an overall architectural comparison against standard baselines (Section 5.1) and a detailed performance breakdown of the proposed model across different playing plucking styles (Section 5.2).

5.1. Models vs Baselines

Three deterministic baselines were evaluated alongside the neural networks:

1. **Identity** baseline, which passes the clean input signal completely unaltered;
2. **Gain Matched** baseline, which scales the clean input amplitude to match the root-mean-square (RMS) level of the target saturated audio;
3. **Static Nonlinear** baseline, which applies a memoryless hyperbolic tangent function to simulate instantaneous wave-clipping without temporal or frequency-dependent dynamics, followed by the same RMS normalization as the Gain Matched baseline.

As demonstrated in Table 3, both proposed models **significantly outperform** all baseline configurations across all metrics, confirming that the neural networks successfully learn the nonlinear saturation transformation rather than merely approximating the input signal.

Table 3. *Test Set Evaluation: best Model for each architecture vs Baselines. The best results are highlighted*

Approach	Total Loss	Time Domain		Freq. Domain
	$\mathcal{L}_{\text{total}}$	**ESR**	**DC**	**MRSTFT**
Proposed Models				
LSTM 3	**0.17684**	**0.06124**	**0.00013**	**0.33170**
WaveNet 2	0.34709	0.15802	0.00159	0.61294
Baselines				
Identity	1.63759	1.42146	0.00026	2.21758
Gain Matched	2.00331	2.02168	0.00039	2.43003
Static Nonlin.	2.02604	2.07576	0.00227	2.42604

The *LSTM 3* model achieves the best overall performance, with a Total Loss of 0.177, an ESR of 0.061 and an MRSTFT of 0.332, reductions of **approximately 85%** relative to the identity baseline.

The *WaveNet 2* model, while considerably weaker than the LSTM, still substantially outperforms all baselines, confirming that both architectures capture meaningful aspects of the multiband saturation behaviour.

It is worth noting that the MRSTFT loss accounts for the **largest share of the total loss** in both models, which is expected given the spectrally complex nature of multiband saturation.

The DC loss remains negligible across all models, confirming that no systematic output offset is introduced. Interestingly, the *LSTM 3* achieves a **lower DC loss than the Identity baseline**, demonstrating exceptional stability in maintaining the signal's vertical alignment. This is a remarkable result since neural emulations of non-linear audio saturation frequently introduce subtle waveform asymmetries that shift the signal's mean, as evidenced by the relative elevated DC loss of *WaveNet 2* or even by the Static Nonlinear baseline.

5.2. Plucking Style Comparison

Table 4 presents a breakdown of test set performance by playing technique for both proposed models.

Notably, both the *LSTM 3* and *WaveNet 2* exhibit the **same relative performance across Plucking Styles**, with Muted (MU) achieving the lowest total loss in both cases (0.150 and 0.341 respectively), followed by Picked (PK) and Slap-Thumb (ST), while Fingerstyle (FS) consistently yields the highest loss across both architectures. This agreement between two structurally different models suggests that the observed pattern reflects a genuine property of the data rather than an architectural bias.

The superior performance on Muted notes is likely at-

Table 4. *Test Set Evaluation: Detailed Performance per Model and Plucking Style*

Model / Style	Total Loss	Time Domain		Freq. Domain
	$\mathcal{L}_{\text{total}}$	ESR	DC	MRSTFT
LSTM 3				
Finger Style (FS)	0.19810	0.07623	0.00021	0.36394
Muted (MU)	**0.15032**	0.05621	**0.00006**	**0.27782**
Picked (PK)	0.17314	**0.04898**	0.00012	0.33575
Slap-Thumb (ST)	0.18257	0.06147	0.00012	0.34421
WaveNet 2				
Finger Style (FS)	0.35520	0.15694	0.00213	0.63191
Muted (MU)	**0.34142**	0.20894	**0.00113**	**0.54951**
Picked (PK)	0.34166	**0.12781**	0.00173	0.63105
Slap-Thumb (ST)	0.34555	0.13290	0.00116	0.63473

tributable to their simpler temporal envelope and reduced sustain (shorter sound), which limits the complexity of the nonlinear transformation the model must learn. Fingerstyle notes, on the other hand, exhibit longer decay and richer harmonic content during the sustain phase, requiring the model to track complex spectral dynamics over longer time windows, which is a more demanding task for both recurrent and convolutional architectures.

6. Signal Comparison

With the predicted model outputs (from *LSTM 3* and *WaveNet 2*) and the target sound (from the Saturn 2 VST) we can compare them across multiple domains.

Figure 8 presents a waveform comparison between the Saturn 2 target output and both model predictions over 50 ms. The *LSTM 3* predicted waveform tracks the target closely across the full time window, with the error signal being consistently minimal, exhibiting no systematic bias or drift. In contrast, the *WaveNet 2* output shows light but noticeable deviations throughout the waveform.

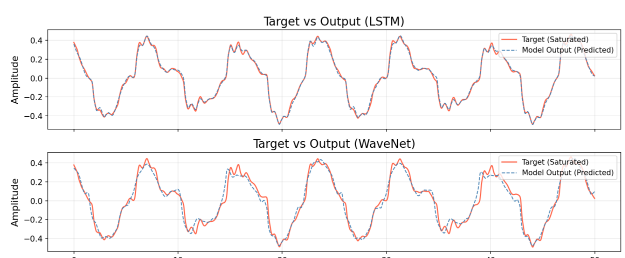

Figure 8. *Target vs. Predicted waveform comparison over 50 ms. The LSTM 3 error signal (top) remains minimal throughout, while WaveNet 2 exhibits larger residual deviations.*

The magnitude spectrum (Figure 9) shows a tight alignment with the Saturn 2 target across the audible range, successfully preserving the overall spectral envelope despite minor harmonic deviations above 2 kHz. This boundary

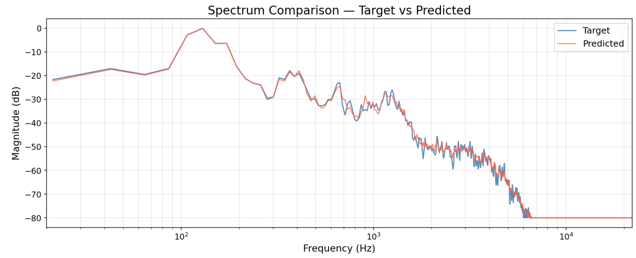

Figure 9. *Magnitude spectrum comparison between the Saturn 2 target and LSTM 3 prediction.*

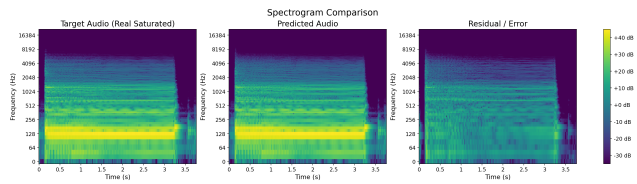

Figure 10. *Spectrogram comparison showing the Saturn 2 target (left), the neural network prediction (center), and the residual error (right).*

is further detailed by the time-frequency spectrogram comparison (Figure 10), where the residual error (right) remains exceptionally low throughout the note's duration. The minor, well-distributed background noise visible above 1 kHz aligns with the multi-resolution spectral loss (MRSTFT) values recorded during training, confirming that the model faithfully tracks the dynamic timbral evolution without introducing disruptive spectral artifacts.

These observations are further supported by informal listening tests[3]. The *WaveNet 2* output exhibited perceptible timbral discrepancies relative to the target. The *LSTM 3* output, by contrast, was perceptually indistinguishable from the target under normal listening conditions.

In the remaining LSTM versions, the difference was noticeable, as they delivered generally poorer results, meanwhile, the gap between *WaveNet 2* and *LSTM 3* was minimal.

7. Multiband vs Single Band

Using the *LSTM 3* model, we also compared the performance of multiband and single-band saturation presets. For the single-band configuration, the parameters from Band 2 were applied uniformly across the entire frequency spectrum, resulting in a more cohesive and uniform saturation effect. Theoretically, the model should face fewer difficulties learning this behavior, as the absence of crossover networks eliminates phase alignment complexities and spectral

[3]Audio examples are available at https://joao-canais.github.io/Black-Box-Modelling-of-Multiband-Saturation/

variations.

Table 5. *Multiband vs Single Band: Test Set Evaluation. The best results are highlighted.*

Approach	Total Loss	Time Domain		Freq. Domain
	$\mathcal{L}_{\text{total}}$	ESR	DC	MRSTFT
Multiband	0.17684	0.06124	0.00013	0.33170
Single Band	**0.16395**	**0.05259**	**0.00012**	**0.31172**

As shown in Table 5, the single-band approach achieved lower loss values across all metrics. Although the performance margin is narrow, these results support the hypothesis that emulating multiband processing introduces a higher degree of complexity for the network.

The fact that the discrepancies between the two approaches are less pronounced than initially expected can likely be attributed to the specific configuration of the multiband preset. Because the same saturation style (*Power Amp*) was applied across all four bands, varying only in the way that it's applied (localized parameters), the underlying nonlinear behavior remained relatively consistent.

Figure 11. *Target vs. Predicted waveform comparison for the single band preset over 50 ms (top) and the residual error (bottom).*

8. Conclusions

This work has compared the performance of two types of neural networks for offline emulation of multiband saturation. The proposed bidirectional LSTM model achieves the highest emulation accuracy compared to the Wavenet model.

The evaluation across Plucking Styles revealed identical performance patterns for both architectures. Muted notes achieved the highest accuracy due to their simple envelopes, while Fingerstyle proved to be the most challenging due to its complex spectral decay. Furthermore, a comparison with a single-band configuration confirmed that emulating multiband processing introduces higher complexity.

Future work should focus on real-time implementation with latency compensation, as well as dataset expansion to improve model generalization beyond bass guitar signals. Furthermore, given the flexibility of the plugin, expanding the training to new and more complex multiband saturation presets will be essential to characterize the relationship and relative difficulty between multiband and single-band emulation.

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